

Ch3_Data Product

What is a data product ?

1. Definition: **Solution** with data as the core component
2. Importance: Enable DDDM
3. Example: Dashboards, recommendation engines, predictive models...
 - Dashboard: 面向决策者的数据产品, 提供可视化洞察
 - Recommendation Engine: 面向消费者或终端用户的数据产品, 个性化建议



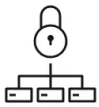
Discoverable

Stakeholders should be able to easily discover and find the right data product for their use case.



Understandable

A data product should include clear [metadata](#) and be structured according to specific business domains, enabling data consumers and domain teams to interpret and apply the information effectively.



Secure

A data product should have access controls and [security](#) measures in place to ensure that only authorized users can access the data while maintaining [compliance](#).



Reusable

A well-designed data product is built from modular components that can be repurposed to create new data products or derivative insights, increasing efficiency and reducing redundant efforts.

4. Advantage:

- Internal: Support organizational decision
- External: Providing value to customers (fulfill their needs)
- 数据产品如何为用户创造价值? -- Value Creation in Data Products
 - 客户画像 (Customer Profile) && 价值地图 (Value Map)



Value map:

Products & services: Data, insights, visualizations
Pain relievers: How the product addresses challenges
Gain creators: How the product delivers benefits

Customer profile:

Jobs: Decisions to be made
Pains: Risks, frustrations, challenges
Gains: Desired insights, benefits

产品角度出发，展示如何回应用户画像

- **Products & Services (产品与服务)**：你提供的数据、分析、可视化工具等。
- **Pain Relievers (痛点缓解器)**：产品如何解决用户的痛点，比如“自动清洗数据”、“提供解释性分析”、“简化操作流程”。
- **Gain Creators (收益创造器)**：产品如何带来额外好处

理解用户需求

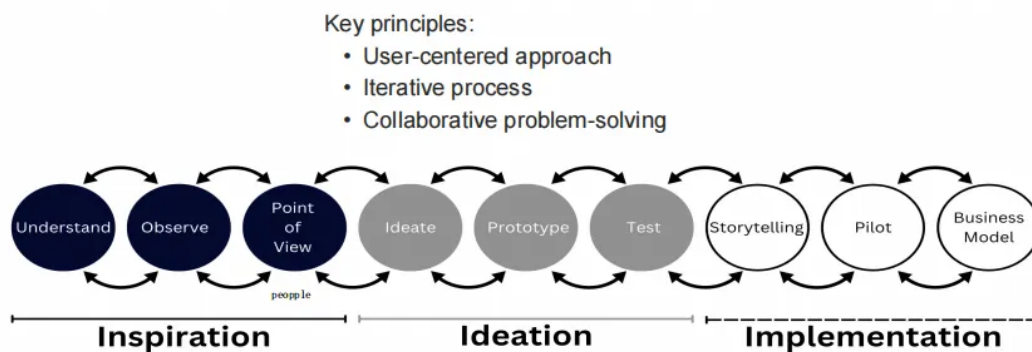
- **Jobs (任务)**：用户需要完成的决策或行动，比如“选择合适的学习资源”或“判断某个数据是否有用”。
- **Pains (痛点)**：用户在完成任务时遇到的困难，比如“信息太多太杂”、“数据不透明”...
- **Gains (收益)**：用户希望获得的好处，比如“清晰的洞察”、“节省时间”、“提升决策质量”。

分析用户需求 --> 价值创造的起点：识别用户任务、痛点、期望

数据产品 --> 对用户需求的理解转化为具体的产品功能。数据本身不是产品，只有当它被组织、封装、并用于解决用户问题时，才成为数据产品

Design thinking for data products

设计思维”在数据产品开发中的应用流程



将传统设计思维的三个阶段，重新框架为适用于数据产品的开发路径。

1. 循环式，可回溯，持续改进
2. 用户反馈驱动
3. 协作解决问题：一种通过团队合作来解决问题的方法，通常涉及多个人员的参与和贡献。

Inspiration

- **Understand (理解)**：收集背景信息，明确**用户是谁**，他们的目标和挑战。
- **Observe (观察)**：通过访谈、行为观察、数据分析等方式，发现用户真实行为和隐性需求。
- **Point of View (观点定义)**：将观察结果转化为明确的问题陈述，通常是“用户需要一个……因为……”

Ideation

- **Ideate (构思)**：头脑风暴各种可能的解决方案，不设限。
- **Prototype (原型)**：将想法变成可测试的模型
- **Test (测试)**：让真实用户使用原型，收集反馈，发现问题。

Implementation (实施)

- **Storytelling (讲述)**：清晰表达产品的设计逻辑、用户价值和使用场景。
- **Pilot (试点)**：在小范围内部署产品，观察实际效果。
- **Business Model (商业模式)**：思考产品的持续运营方式，比如数据更新机制、用户增长策略、价值闭环。

Inspiration

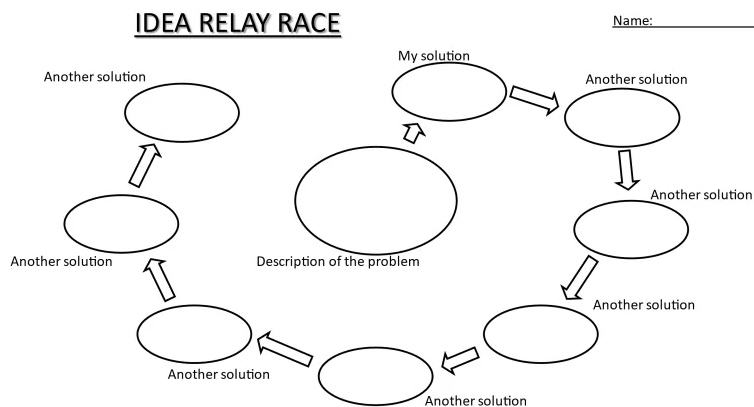
- Empathy, Understanding user **needs and perspectives**
- Techniques
 - **User Interviews (用户访谈)**：直接与目标用户对话，了解他们的动机、障碍、期望。
 - **Shadowing (跟踪观察)**：像影子一样陪伴用户，记录他们的行为和决策过程。
 - **Contextual Inquiry (情境访谈)**：在用户实际操作过程中进行访谈，边看边问，更贴近真实使用场景。
 - **Analogous Inspiration (类比启发)** – 从其他领域寻找相似问题的解决方式。例如：从游戏设计中借鉴学习激励机制
 - 属于Inspiration：在启发阶段，我们的目标是拓展对问题空间的理解，类比启发通过从其他领域寻找相似问题的解决方式，帮助我们重新框架问题。
 - 属于Ideation：类比启发成为创意的跳板
 - **Extremes and Mainstreams (极端与主流用户)**：同时关注典型用户和边缘用户，发现更广泛的需求空间。
- **Comparison – Interview vs. Contextual Inquiry vs. Observation**
 - It happens in the user's natural environment (e.g., classroom, app usage session, workplace).
 - The researcher **observes the user doing real tasks**, not just talking about them.

- While observing, the researcher **asks questions in context**

Method	Focus	Researcher Role	User Role
Interview	Thoughts, opinions, memories	Listener	Narrator
Observation	Behavior, workflow	Silent observer	Actor
Contextual Inquiry	Behavior + reasoning	Observer + questioner	Actor + explainer

Ideation

- Generate diverse solution ideas
- Brainstorming (头脑风暴)** 快速提出大量想法，不设定、不批评，鼓励自由联想。
 - IDEA RELAY RACE” (创意接力赛) -> 渐进式头脑风暴



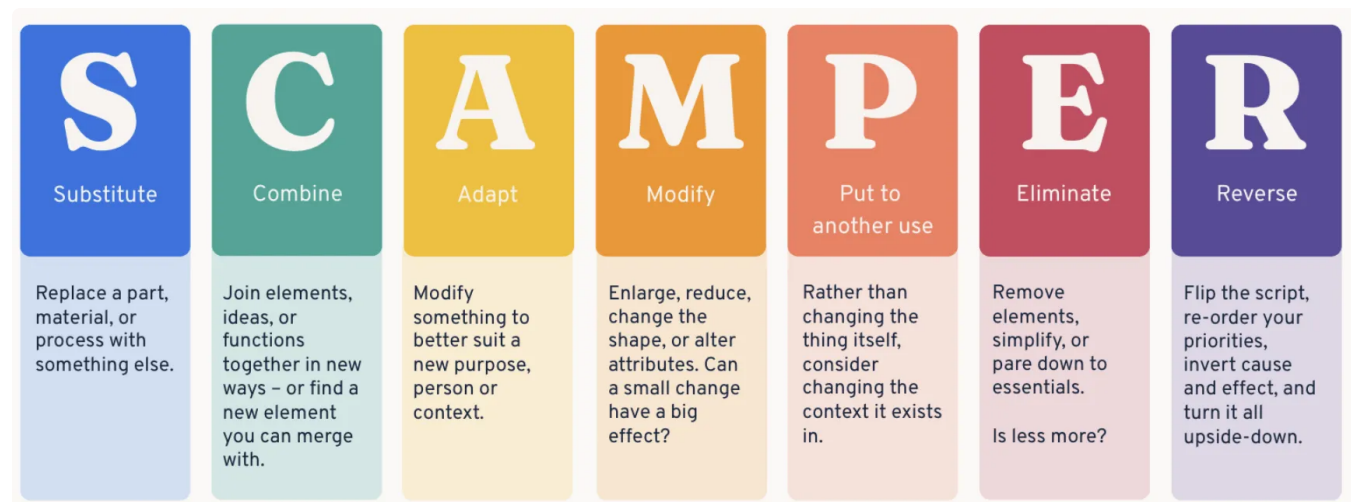
- Worst Possible Idea (最糟糕的点子)** 故意提出荒谬或失败的方案，反向激发创新思维。

Worst Possible Idea is an ideation method where team members purposefully seek the worst solutions in ideation sessions. The “inverted” search process relaxes them, boosts their confidence and stokes their creativity so they can examine these ideas, challenge their assumptions and gain insights towards great ideas.

- Come up with as many **bad ideas** as we can.
- List *all* the properties of those terrible ideas.
- List what *makes* the worst of these so very bad.
- Search for the *opposite* of the worst attribute.
- Consider *substituting* something else in for the worst attribute.
- Mix and match various awful ideas to see what happens.

- SCAMPER 法** (Substitute, Combine, Adapt, Modify, Put to other uses, Eliminate, Reverse)

It encourages us to **re-examine** existing products, services, or processes through seven different lenses, prompting a re-evaluation of what we thought was set in stone



代替 产品或流程中的哪些元素可以被替换，而不会降低其价值

结合 指将两个或多个产品或流程的元素合并在一起，以创造新产品或增强功能。

适应 思考如何调整或重新利用产品或流程中的元素，使其发挥其他功能

- 我们可以采用其他领域或行业的哪些现有解决方案来应对我们的挑战？
- 我们如何修改产品或服务的某个方面来满足不同的目的或受众？
- 我们怎样才能将当前的局限性转化为优势或机遇？

调整 鼓励调整元素以改进产品或流程，重点关注形式、形状或属性的变化

另作他用 如何将产品或流程重新利用或用于其最初设计用途以外的场景

排除 侧重于简化，通过移除不必要或低效的元素，甚至是那些看似重要的元素。

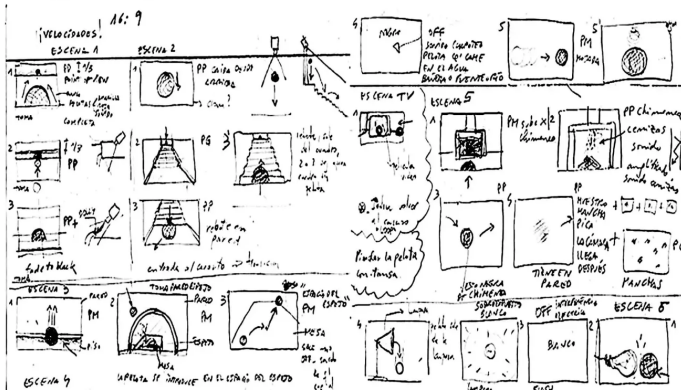
R反转（或R重排） 从当前做法的反面出发，或者重新排列各个部分，看看是否有更好的解决方案。通过讨论来扭转局面.....

- 如果我们采取与当前相反的做法，我们的产品、服务或流程会是什么样子？
- 我们能否交换操作顺序、服务流程或功能层次以获得有益的结果？
- 如果我们扭转目标受众的期望，我们的方法或产品会如何改变？

<https://www.bitesizelearning.co.uk/resources/scamper-model-creativity>

- Tools: Mind mapping/ storyboarding – 视觉框架，用于有效地规划和传达想法

Storyboarding for Business



Implementation

Prototyping: Building tangible representations

Types of prototypes:

- Low-fidelity: Sketches, paper prototypes
- High-fidelity: Interactive digital mockups

Testing:

- User feedback sessions
- A/B testing
- Usability studies

原型类型 (Types of Prototypes)

原型是将想法变成“可看、可触、可体验”的形式，分为两种：

Low-fidelity (低保真原型)

- 形式：草图、纸面界面、流程图
- 特点：快速、便宜、易修改
- 用途：用于早期构思阶段，验证功能逻辑和用户流程

High-fidelity (高保真原型)

- 形式：可交互的数字模型
- 特点：接近真实产品，包含视觉设计和部分功能
- 用途：用于用户测试阶段，收集更真实的反馈

A/B Testing (A/B 测试)

定义：将用户随机分成两组，分别使用两个不同版本的产品（A版和B版），比较哪一个效果更好

用途：测试设计细节对用户行为的影响，如按钮颜色、功能顺序、推荐算法

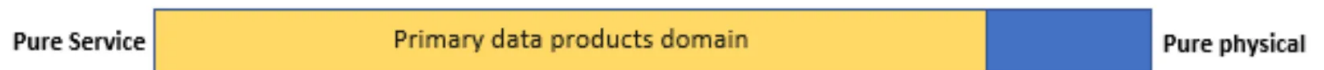
关键指标：点击率、完成率、停留时间等

Usability studies, or usability testing, is a user-centered research method where real users are observed performing tasks on a product to assess its ease of use, identify design flaws, and uncover pain points 通过观察真实用户在产品上执行的任务来评估产品的易用性、识别设计缺陷并发现痛点

Data product canvas overview

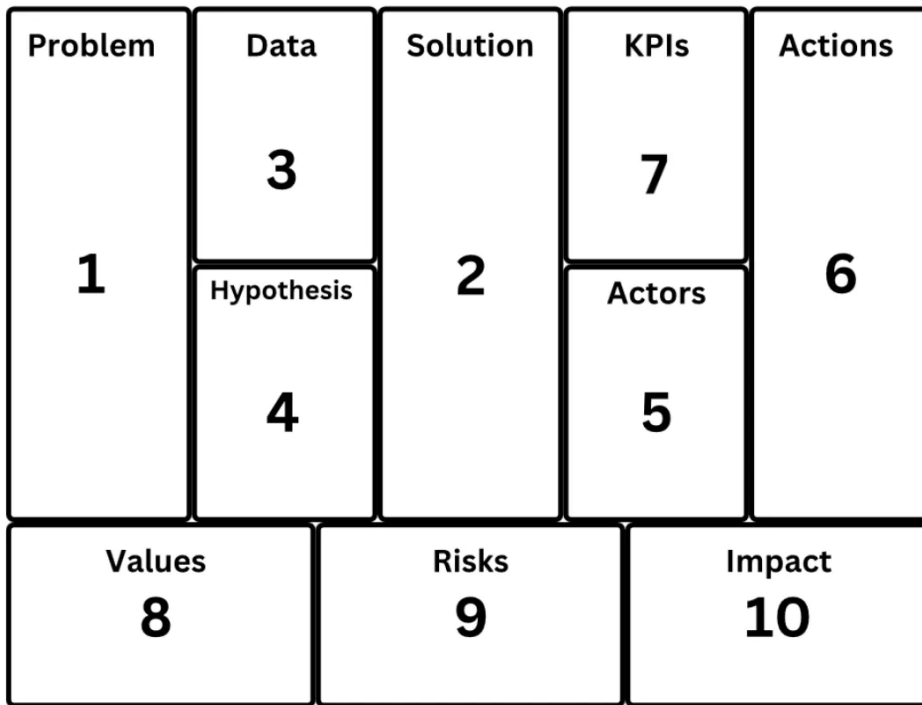
The **Data Product Canvas** is a **design tool** used to collaboratively define the key elements of a data product.

数据产品的定位光谱



- Pure service – 纯服务型
 - 完全以服务为核心，数据只是支撑
 - 咨询服务、人工客服
- Primary Data Products Domain（核心数据产品区）
 - 数据是产品的核心
 - 仪表盘、推荐系统、预测模型
- Pure Physical（纯物理型）
 - 实体产品为主，数据只是附加
 - 智能手环、IoT设备、硬件终端

数据产品画布（Data Product Canvas） – Before produce



Follow the numbers when filling the canvas

- **Problem** – Challenge they face?
 - Importance of clear problem statements
 - Use "5 Whys" technique for root cause analysis

Questions to ask:

- What is the problem?
- Why is it a problem?
- Whose problem is it?

- **Solution**

Defining the simplest effective solution

- Avoiding over-engineering

Over-engineering is the act of designing a solution to a problem that is unnecessarily complex, providing more features, robustness, or complexity than is needed for its intended use

- **Data** – What data is available or needed?

Data mapping considerations:

- Sources
- Quality
- Access
- Transformation requirements
- ...

What do data maps do?

- Move data into the right fields in a database or software.
- ☑ Correct and standardize data formats so it's all the same.
- ① Prevent errors or data discrepancies.

数据映射是将一个数据库的字段匹配到另一个数据库的过程

数据帮助我们确认问题是否真实存在、影响多大、涉及哪些用户。

数据是产品功能的基础，比如推荐算法、图像识别、个性化词库。

- **Hypothesis** – What assumption connects data to the problem?

假说是对“数据如何解决问题”的**有逻辑的预测**，它让设计变得有目标

假说回答：“如果我们用这些数据做某件事，会不会改善问题？”

假说是后续 KPI 和测试设计的基础。

Formulating testable hypotheses

- Example: "Implementing solution X will reduce churn by Y%"

一个**假说 (hypothesis)** 是对某个现象或结果的**有逻辑依据的预测**，而“可验证”意味着它可以通过数据或实验来**证实或否定**。

干预措施 (Intervention)	做了什么?	实施方案 X
预期结果 (Expected Outcome)	会发生什么?	客户流失率降低
量化指标 (Measurable Metric)	如何衡量?	降低 Y% 的流失率

- **Actors** – Who are the users and stakeholders?

Identifying key stakeholders:

- Users
- Decision-makers
- Data owners
- IT support
- ..

- **Actions** – What do users do with the product?

Defining actions enabled by the product

- Example: Personalized retention campaigns

用户因为使用你的数据产品而能够采取的具体行动

Personalized retention campaigns 利用客户数据 **leverage customer data** 来创建有针对性的相关营销信息、优惠和体验，以培养信任、忠诚度并提高终身价值

更倾向于：产品为了促使用户采取某种行为（如继续使用、不流失）而主动采取的策略或功能。

- **KPIs** – What metrics measure success?

Selecting relevant performance indicators:

- Technical KPIs: Accuracy, response time
- Business KPIs: Revenue impact, user adoption

Technical KPIs

技术层面的表现，通常用于评估代码质量、系统效率或算法性能：

- **Accuracy（准确率）**
- **Response Time（响应时间）** 衡量系统对请求的处理速度。比如一个网页加载时间、API调用的延迟，都是关键的用户体验指标。

Business KPIs

关注的是技术成果是否带来了**实际的业务价值或用户行为变化**：

- **Revenue Impact（收入影响）** 比如是否提高了销售额，或一个优化后的流程是否节省了成本。
- **User Adoption（用户采纳率）** 衡量有多少用户开始使用某个新功能或产品。比如一个新上线的移动端功能是否吸引了更多活跃用户。
- **Values**

Estimating problem magnitude: Potential cost savings or revenue gains

估算问题规模：潜在的成本节省 或 收入增长

这部分强调的是：在项目启动或技术方案设计前，先要评估“问题有多大”，也就是：

- 如果我们解决了这个问题，会带来多少好处？
- 如果我们不解决，会损失多少？

这是一种**价值导向的思维方式**，目的是量化问题的商业影响。

Cost savings	成本节省：例如自动化流程减少人工成本，优化资源使用	间接价值
Revenue gains	收入增长：例如新功能吸引更多用户，提升转化率	直接价值

- **Risks**

Identifying and mitigating risks:

- Data quality issues
- Privacy concerns
- User adoption challenges

Data quality issues 数据质量问题	数据不完整、错误或不一致，可能影响分析结果或模型表现。
Privacy concerns 隐私问题	涉及用户数据的收集与使用，必须符合隐私法规（如 GDPR），否则可能引发法律风险。
User adoption challenges 用户接受度挑战	即使技术方案优秀，如果用户不愿意使用或难以理解，也会影响项目成功。

• Impact

Measuring business impact:

- Short-term vs long-term effects
- Direct and indirect benefits

Short-term vs long-term effects 短期与长期影响	项目是否能带来即时收益，还是需要时间积累才能体现价值。
Direct and indirect benefits 直接与间接收益	例如直接节省成本 vs 间接提升品牌形象或客户满意度。

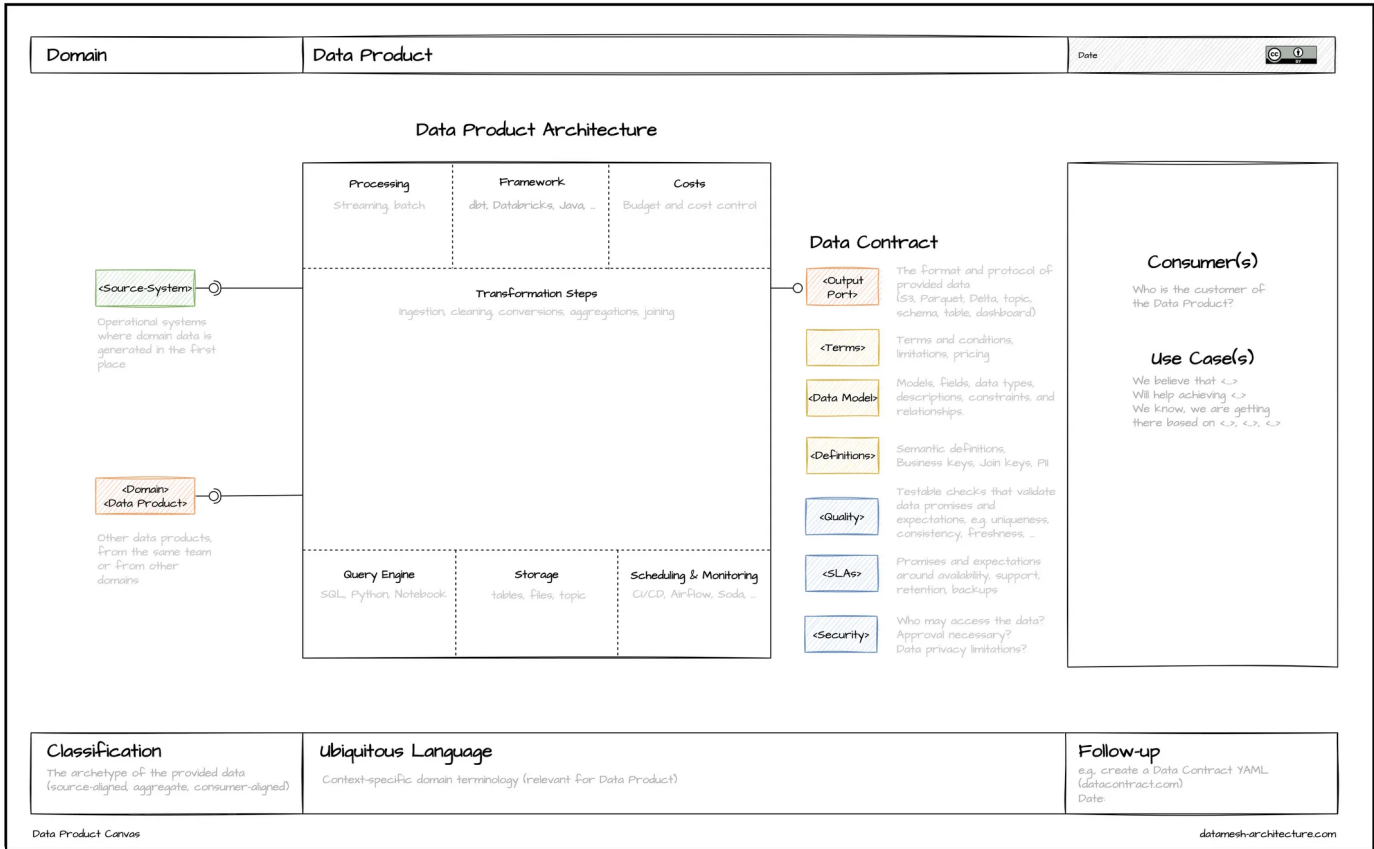
-Risk evaluation-

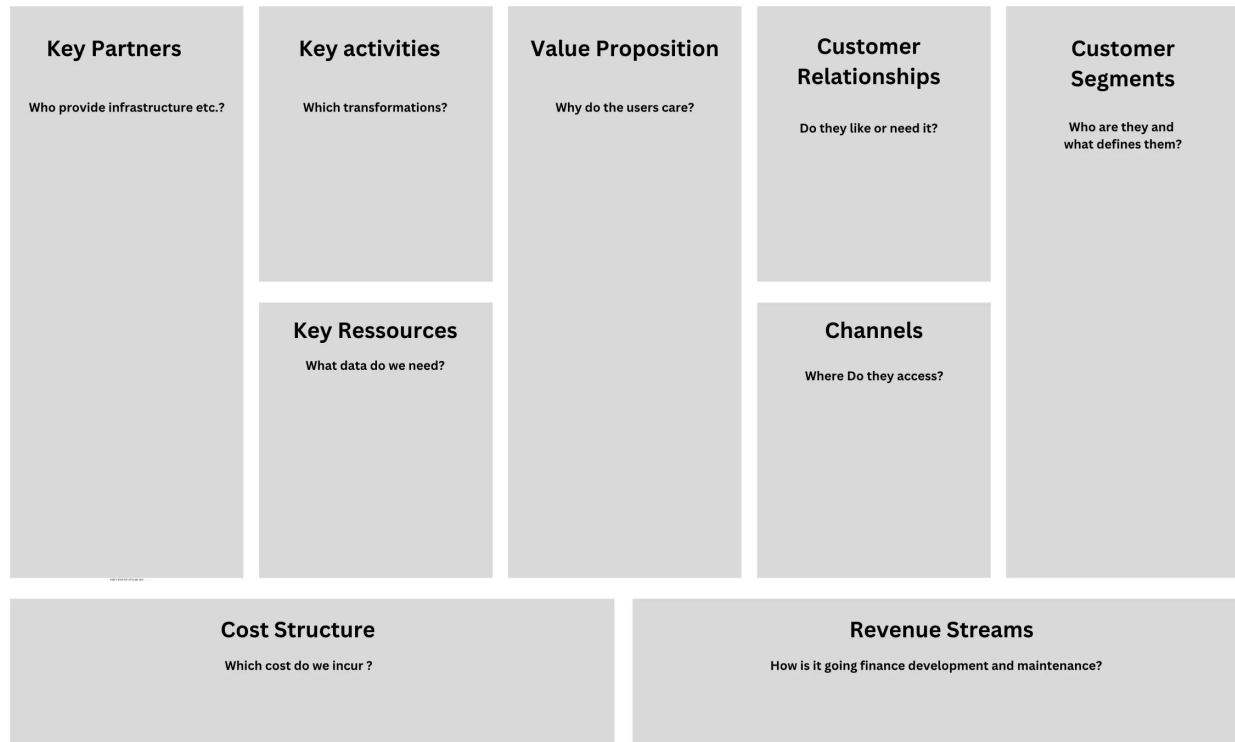
5 4 3 2 1 1 2 3 4 5 Probability Impact		Project Objectives	1 Very Low	2 Low	3 Moderate	4 High	5 Very High
		Cost	Insignificant Increase	< 10% cost increase	10-20% cost increase	20-40% cost increase	>40% cost increase
		Time	Insignificant Increase	< 5% time increase	5-10% time increase	10-20% time increase	>20% time increase
		Scope	Scope decrease barely noticable	Minor areas of scope affected	Major areas of scope affected	Scope reduction unacceptable to sponsor	Project has failed
		Quality	Quality lack barely noticable	Only high-Performance solutions affected	Quality reduction require sponsor approval	Quality reduction unacceptable	Project has failed

Table 3.1: risk assessment matrix

Alternative data product models

Data product canvas is one of the data product model.





Data Value Chain

 **How They Relate**

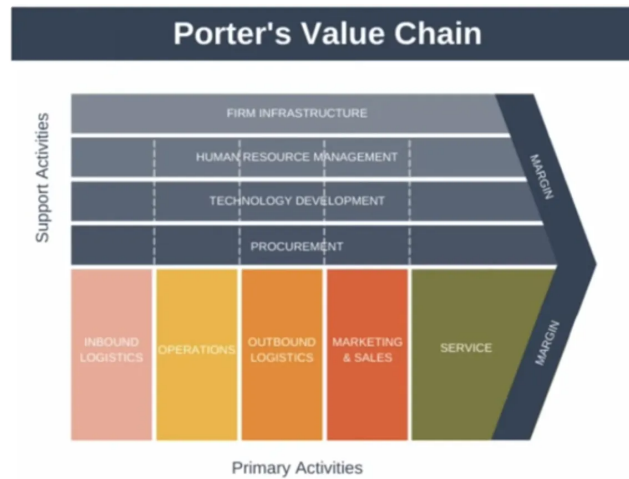
Concept	Role	Relationship
Data Value Chain	Describes the lifecycle of data from raw to impact	Provides the process context
Data Product Model	Describes the structure and governance of a reusable data unit	Implements parts of the value chain as a deliverable

So:

- The **data value chain** is like the **assembly line**
- The **data product model** is like the **finished product blueprint**

They're not the same, but a **good data product model depends on a well-understood value chain.**

- Adapt Porter's value chain for data operation
 - A strategic tool to **map out** internal activities that add value to your customers
 - Value chain – the set of all activities an organization performs to create margin (Value created – cost of creating that value).



– Procurement 是决定成本的关键，因为其决定了用什么资源、从哪里买、以什么价格买，但是 波特的价值链是从“是否直接参与产品或服务的创造”来划分的

– Market&sale: Advertising, partnership etc.

– Service: Maintain/grow product value after selling it

- Steps to create a value chain:
 - Map out sub activities
 - Analyze the sub activities (Value v.s. Cost)
 - Examine the linkages – Change one → influence other

Primary activities, such as [inbound logistics](#), [operations](#), and [marketing and sales](#), are **directly involved in the physical creation, sale, and distribution of a product**, and a company can achieve a cost advantage by performing these activities more efficiently than competitors.

Support activities, like [human resources](#) and [technology development](#), enhance the primary activities and contribute to competitive advantage through **differentiation** by creating unique value or providing the necessary foundation for innovative or superior products.



🧠 What Are Primary Activities?

In Porter's value chain, **primary activities** are the core operations that directly create value:

- **Inbound logistics** (getting materials)
- **Operations** (making the product)
- **Outbound logistics** (delivering it)
- **Marketing & Sales**
- **Service** (after-sale support)

These are contrasted with **support activities** like HR, tech, and procurement.

💰 Why Are Primary Activities Linked to Cost Advantage?

Because they involve **direct handling of inputs, processes, and outputs**, they're often the **biggest cost centers**:

- Efficient operations → lower production cost
- Streamlined logistics → cheaper delivery
- Lean service → reduced support overhead

So when firms optimize these, they can **undercut competitors** on price.

✳️ That's why textbooks often say:

"Primary activities are the main source of cost advantage."

主要活动（Primary Activities）并不局限于成本优势，但它们更容易成为成本优势的来源。

例子：Tiffany's Packaging: Differentiation Through Primary Activity

Tiffany's iconic blue box is **not about cost advantage**—it's about **emotional impact, brand identity, and perceived value**. It happens during the **sales process**, which is a **primary activity** in Porter's value chain.

Why It's Differentiation:

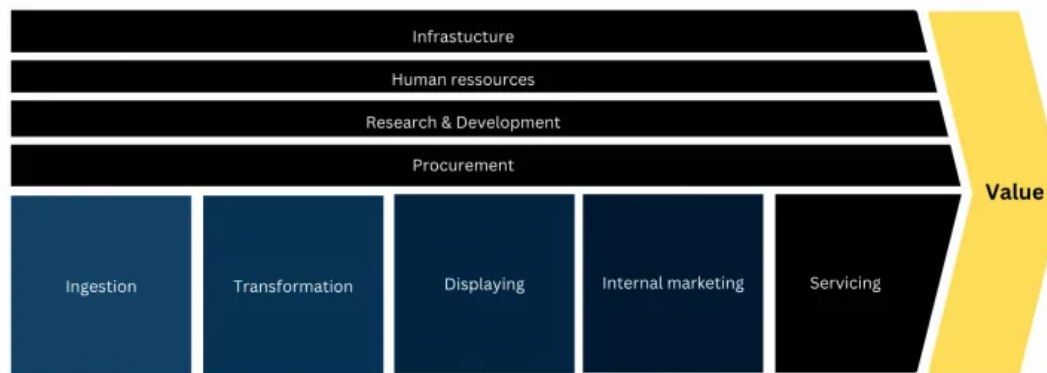
- **Creates emotional resonance**: The packaging signals luxury, care, and exclusivity.
- **Enhances perceived value**: The same product feels more valuable when wrapped in the Tiffany experience.
- **Builds brand memory**: Customers remember the unboxing, not just the product.

Tiffany's packaging is part of **sales execution**, not just branding. It's how the product is **delivered and experienced**, not just advertised.

Primary activities: Core data processing steps

Supporting activities: Enabling functions

Goal: Understand and optimize data flow and value creation



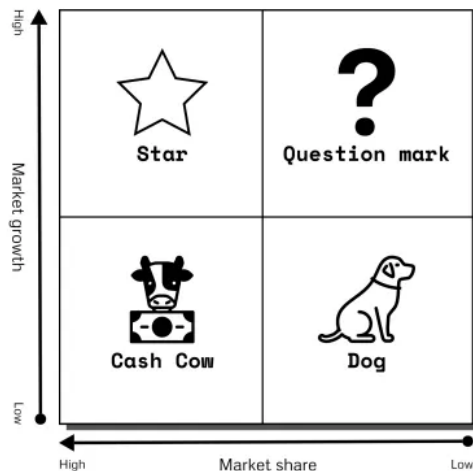
- 数据产品生命周期五阶段
 - **Ingestion** – Data collection and import
 - 从不同来源（如数据库、API、传感器）收集原始数据并导入系统。
 - **Transformation** – Cleaning, integrating, and modeling data
 - 对数据进行清洗、整合和建模，使其结构化、可分析。
 - **Displaying** – Visualizations and reporting.
 - 通过图表、仪表盘或报告将数据呈现给用户。
 - **Internal Marketing** – Promoting the data product within the organization to encourage adoption.
 - 在组织内部推广数据产品，鼓励使用和反馈。
 - **Servicing** – User support and maintenance
 - 提供用户支持、修复问题，并持续维护数据产品。
- Support activities
 - **Infrastructure** – Data storage, processing systems
 - 用于存储、处理和管理数据的系统与平台，如数据库、云计算
 - **Human & resources** – Talent acquisition and development
 - 招聘、培训并留住数据相关人才
 - **Research & Development (研发)** – Developing new data techniques, algorithms, and technologies to improve capabilities.
 - **Procurement** – Data sourcing and governance
 - Acquiring external data sources and ensuring proper governance, licensing, and compliance.
 - 获取外部数据资源，并确保治理、授权和合规性。

Ingestion（数据摄取）	Procurement（采购）
数据收集与导入	数据采购与治理
主要活动（Primary）	支持活动（Support）
如何将数据技术性地导入系统	从哪里、以什么条件合法获取数据资源
实现数据流入，供后续处理	确保数据来源合法、合规、性价比高

Data product portfolio Management

The complete set of all data products offered by an organization

(Data) Product Portfolio



Market penetration / usage:

- Adoption rates
- User engagement metrics

Growth potential / strategic impact:

- Alignment with business objectives
- Potential for scalability or new use cases

- **Market Penetration / Usage:**
 - 采用率（adoption rate）
 - 用户活跃度（engagement metrics）
- **Growth Potential / Strategic Impact:**
 - 是否与业务目标一致
 - 是否具备扩展性或新用途

波士顿矩阵（BCG Matrix）是由布鲁斯·亨德森在1970年为波士顿咨询公司设计的战略分析工具，旨在通过分析产品或业务在市场增长率（衡量市场吸引力）和相对市场份额（衡量企业实力）两个维度上的表现

The BCG (Boston Consulting Group) Matrix is one of the most popular portfolio analysis methods.

The **horizontal** axis -> **Market share**, using relative market share => 体现竞争优势.

- 相对市场份额：一个公司在某个市场中所占的份额与其**最大竞争对手**的市场份额之比。

BCG矩阵的核心假设是：

高市场份额 = 高规模经济 = 高利润潜力

但这个“高”必须是**相对于竞争对手**才有意义。因为：

- 如果你是市场第一，你可能拥有更低的单位成本、更强的议价能力
 - 如果你只是市场第三，即使绝对份额不低，也可能没有战略优势
- ✦ 所以，**相对市场份额更能反映你的竞争地位和盈利潜力**

The **vertical** axis -> the growth rate of a product and its potential to grow in a particular market.

- 所处市场或应用领域的**增长潜力**

维度	含义
横轴： Market Share (市场份额)	某个数据产品在组织或市场中的占有率
纵轴： Market Growth (市场增长率)	所处市场或应用领域的增长潜力

<https://corporatefinanceinstitute.com/resources/management/boston-consulting-group-bcg-matrix/>

Balancing investment across quadrants:

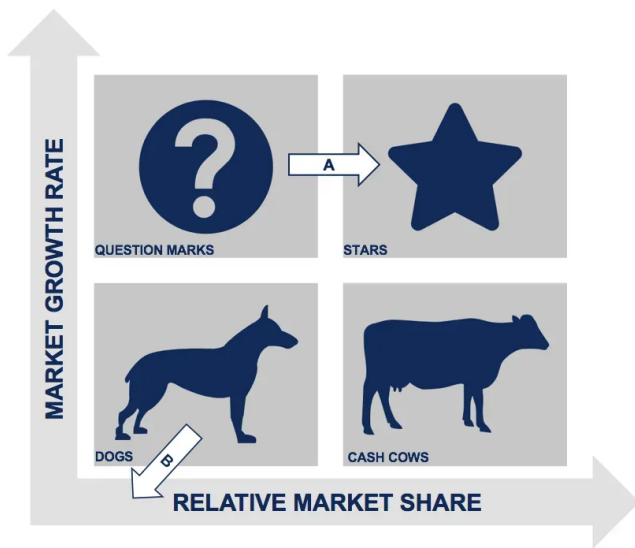
- Stars: High usage, high impact - Invest for growth
- Question marks: Low usage, high impact - Evaluate potential
- Cash cows: High usage, low impact - Maintain efficiency
- Dogs: Low usage, low impact - Consider retiring

Product lifecycle considerations:

- Development
- Launch
- Growth
- Maturity
- Decline or reinvention

✅ 生命周期阶段：

1. **Development (开发)**：构思与原型设计
2. **Launch (发布)**：上线并推广
3. **Growth (增长)**：用户快速增加，功能扩展
4. **Maturity (成熟)**：稳定使用，优化运营
5. **Decline / Retirement (衰退/退役)**：使用减少，考虑淘汰或重构



Question Marks

- 定义：处于快速增长市场，但市场份额较低的产品。
- 特点：
 - 管理难度高，资源需求大。
 - 需要大量投资来提升市场份额。
 - 投资资金通常来自“现金牛”产品的利润。
- 战略路径：
 - 最理想是转化为“明星产品”（路径 A）。
 - 如果无法成为市场领导者，随着市场增长放缓，可能退化为“瘦狗产品”。

Dogs

- 定义：处于低增长市场，且市场份额较低的产品。
- 特点：
 - 通常能维持自身运营，可能有少量现金流。
 - 无法成为明星产品，增长潜力有限。
- 战略路径：
 - 企业通常会逐步淘汰（路径 B）。
 - 除非该产品对现有产品有互补作用或具备战略竞争价值。

Stars

- 定义：处于快速增长市场，且市场份额较高的产品。
- 特点：
 - 市场领先，竞争优势强。
 - 需要持续投资以保持领先地位和推动增长。

- 同时消耗大量资金，也能产生大量现金流。

- 战略路径：

- 随着市场成熟，成功的明星产品将转化为“现金牛”。

Cash Cows

- 定义：处于低增长市场，但市场份额较高的产品。

- 特点：

- 市场稳定，已完成主要投资。
- 不需要大量追加投入即可维持领先地位。
- 产生大量现金流，用于支持“明星”和“问号”产品。

- 战略路径：

- 企业会“榨取”其利润，尽量少投入，持续收获。

Cash Cow 的收益是“可预测的”，但有限；Star 的收益是“潜在的”，但不确定

维度	🐮 Cash Cow (现金牛)	★ Star (明星产品)
市场增长率	低	高
市场份额	高	高
收益稳定性	✅ 高，持续性强	❌ 不确定，波动性大
投资需求	低，维持即可	高，需要持续投入
风险水平	低，成熟市场	高，竞争激烈、变化快
发展空间	小，趋于饱和	大，但不确定是否能兑现
战略角色	提供现金流，支持其他产品	推动增长，争夺未来市场
生命周期位置	成熟期	成长期或快速扩张期